

MICROECONOMIC PRINCIPLES (MANKIW CHAPTERS 4 & 5)

**C211: Global Economics for
Managers**

Western Governors University



WHAT IS A MARKET?

- A grouping of buyers and sellers
 - [Page 4-1a](#) in your text
- Markets can be highly organized (like the Foreign Exchange Market) or very unorganized (like a yard sale)
- For every buyer there is a seller



Pic found at Lonelyplanet.com

WHAT IS COMPETITION?

- Many buyers and sellers
- No one seller or buyer has market power
- No single buyer or seller can affect the equilibrium market price



Pic found at psychologytoday.com

CONSUMER DEMAND

Constrained Utility Maximization

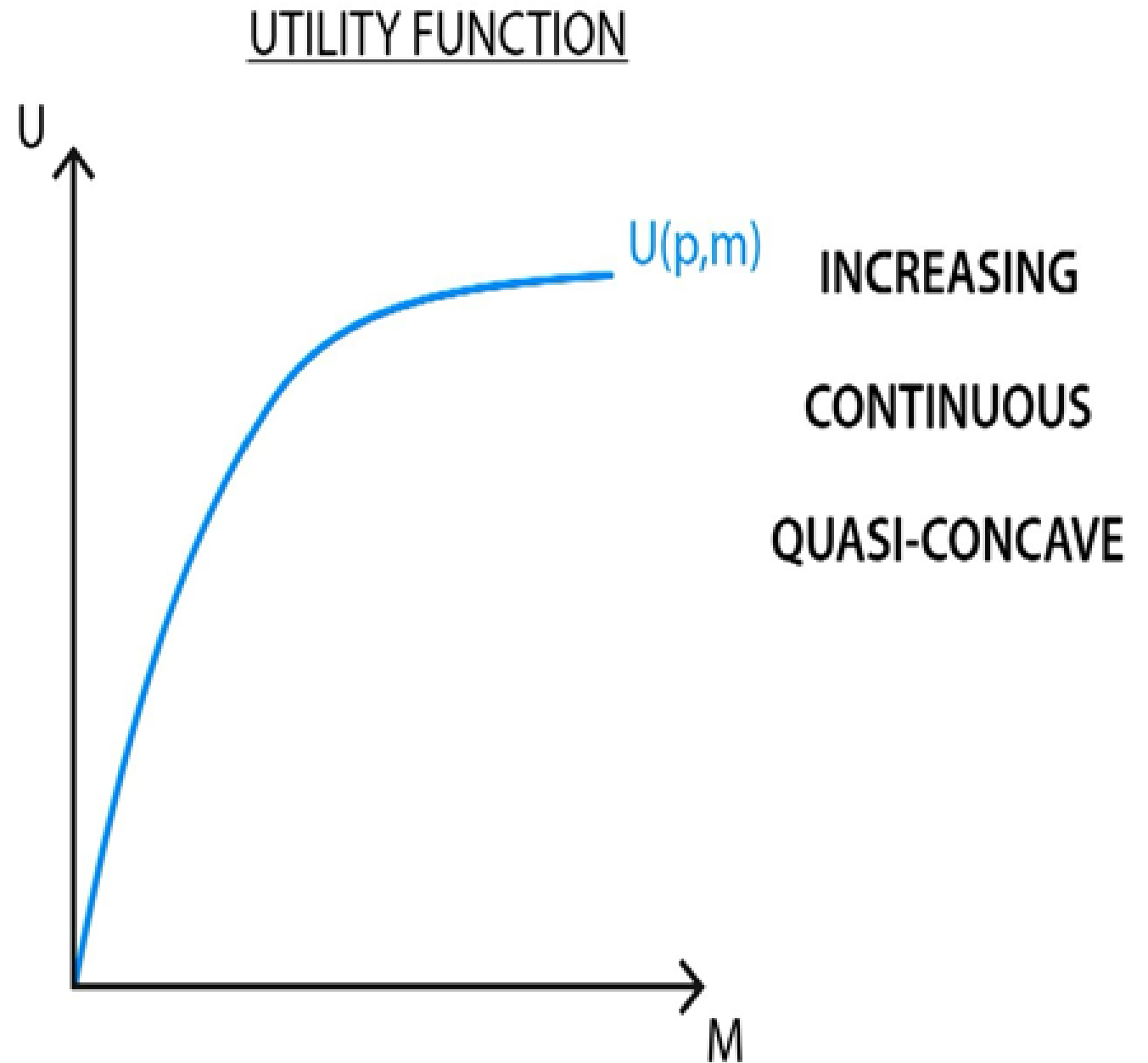
- A Ferrari would make me happiest, but my budget constrains me from having a Ferrari
 - So I buy a Kia instead

Preferences

- Whether I like baseball more than football or football more than baseball; these are my Preferences

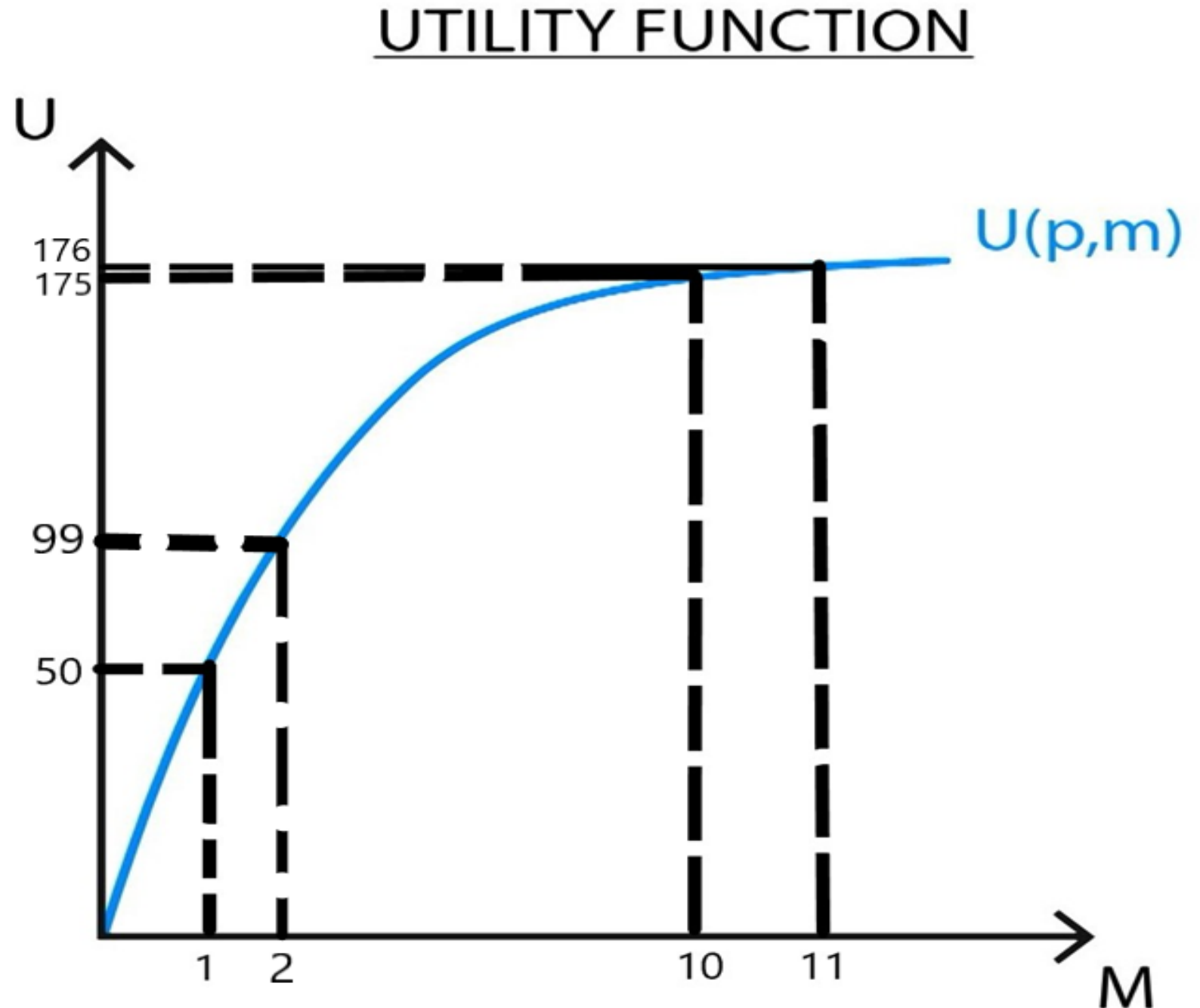
THEORETICAL TOOLS

- We can graph a Utility Function
 - On the Horizontal Axis is M =Mars Bars
 - On the Vertical Axis is U =Utility
- The blue line is my Utility Function
 - In this case, only one good matters to me: Mars Bars
- The more I eat the higher my Utility



THEORETICAL TOOLS

- So My Utility (U) is dependent on the price (p), and the number of Mars Bars I eat (m)
- My Utility rises at first very quickly
 - Goes from 50 to 99
- But then rises more slowly
 - My 11th bar gives me only 1 Util



THEORETICAL TOOLS

So, in economics we assume that the more you consume of a good, the less additional happiness you get from it

This is called Diminishing Marginal Utility

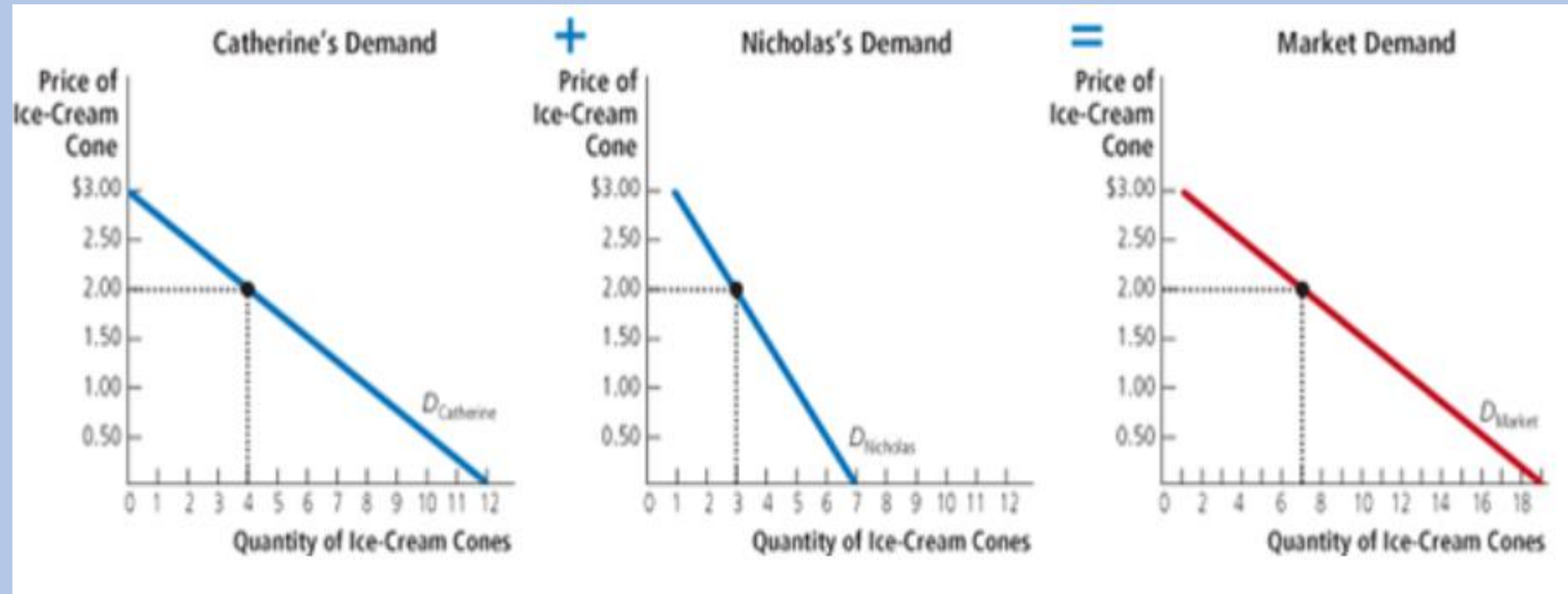


So how much of a good will you consume?

Until the marginal (added) benefit on the next unit you consume equals the marginal cost (the price)

DIMINISHING MARGINAL UTILITY

- From Figure 2 Chapter 4 page 4-2b
- The more ice-cream Catherine consumes, the less she is willing to pay for an additional cone
- This is why demand curves are ALWAYS downward sloping



THE LAW OF DEMAND

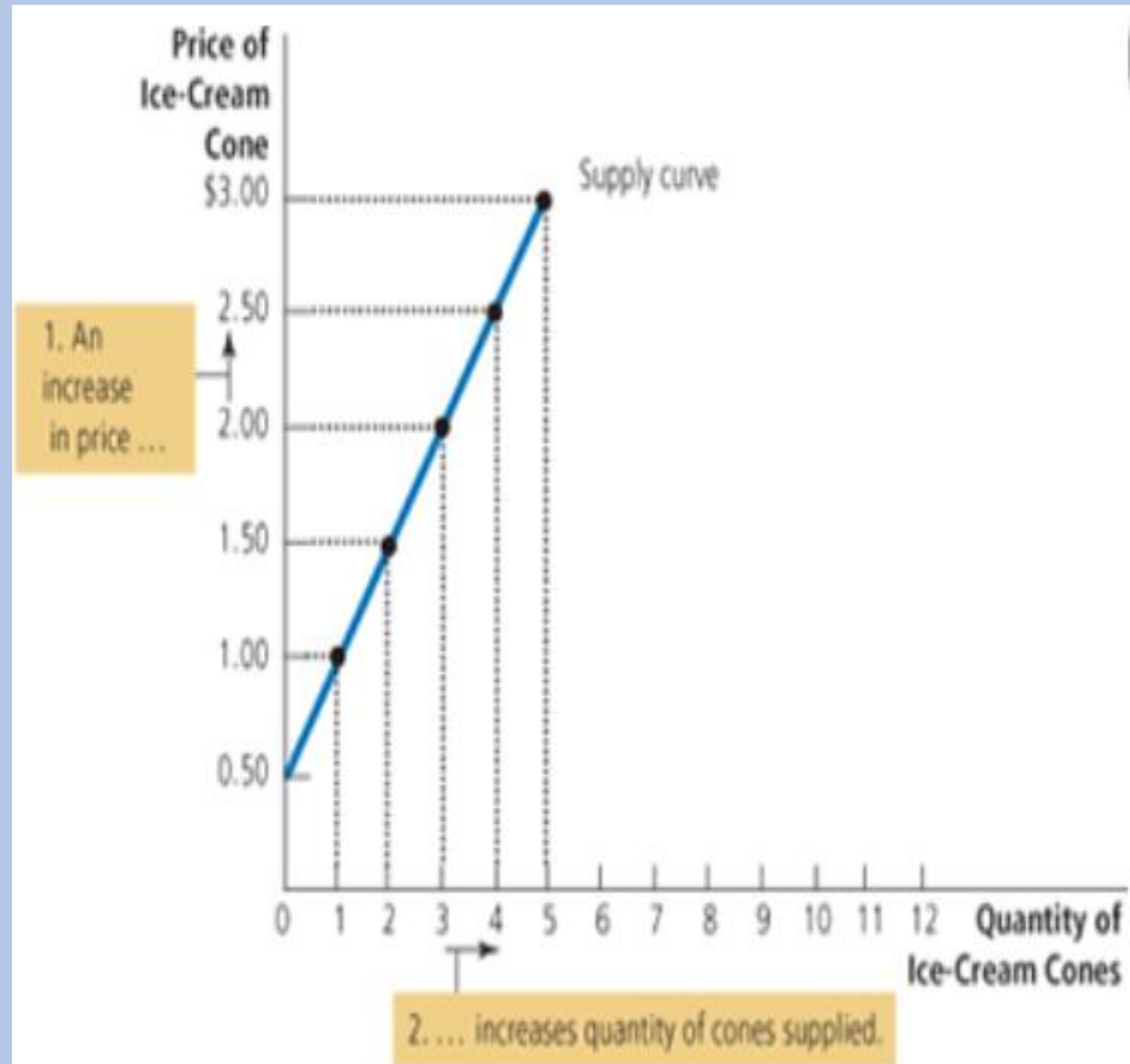
- From Figure 2 Chapter 4 page 4-2b
- When we add Catherine and Nicholas' individual demand (preferences), we get the market demand
- In the Process we have proven The Law of Demand

Price of Ice-Cream Cone	Catherine		Nicholas		Market
\$0.00	12	+	7	=	19 cones
0.50	10		6		16
1.00	8		5		13
1.50	6		4		10
2.00	4		3		7
2.50	2		2		4
3.00	0		1		1

- Law of Demand: As price increases Quantity Demanded falls

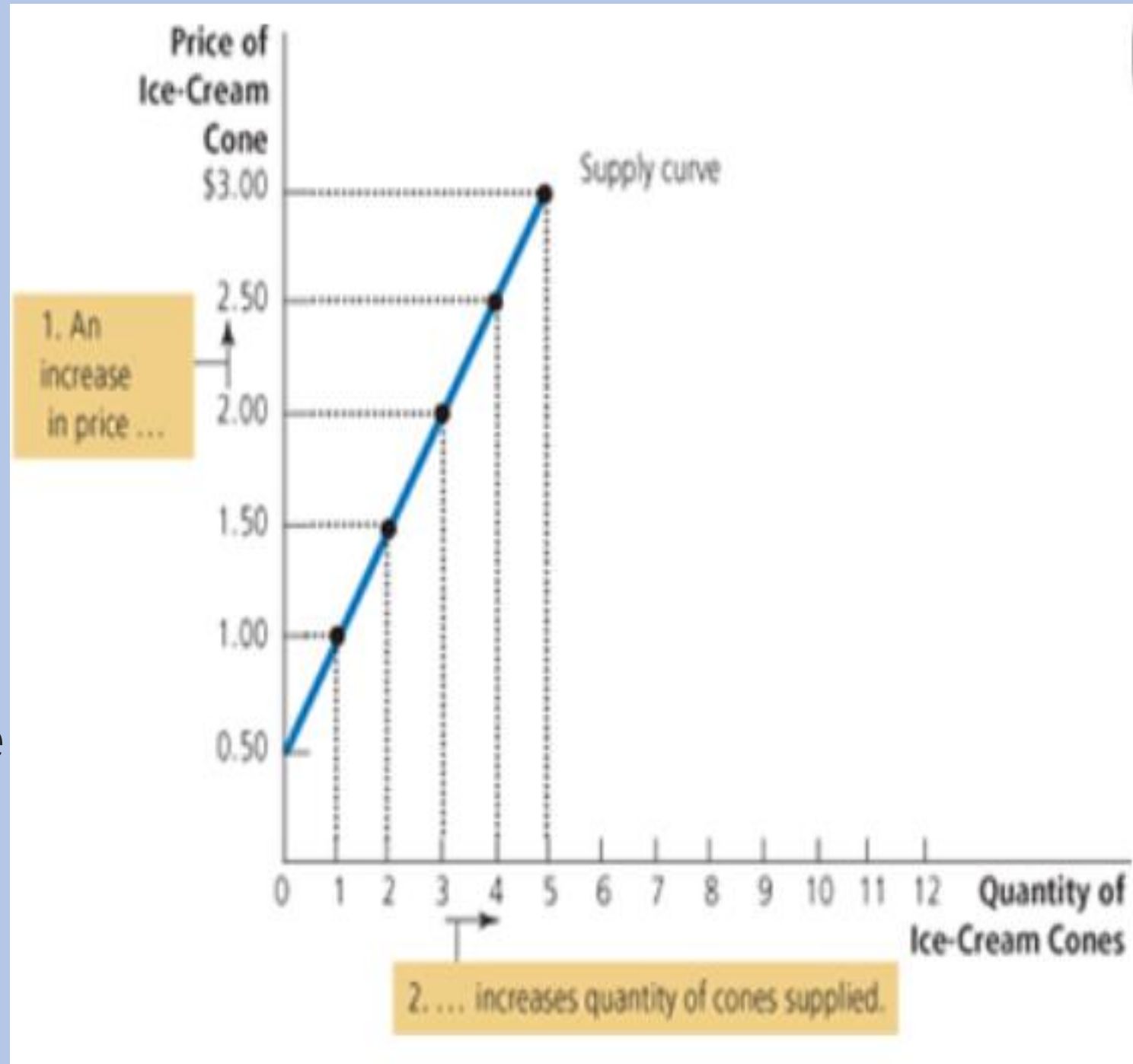
THE LAW OF SUPPLY

- **Supply Curve-** Shows the relationship between the Quantity Supplied and the Price
 - It represents the Marginal Cost of production to the firm
- It is the minimum price the firm can accept
 - Which is why it is sometimes called the **Willingness to Accept Curve**



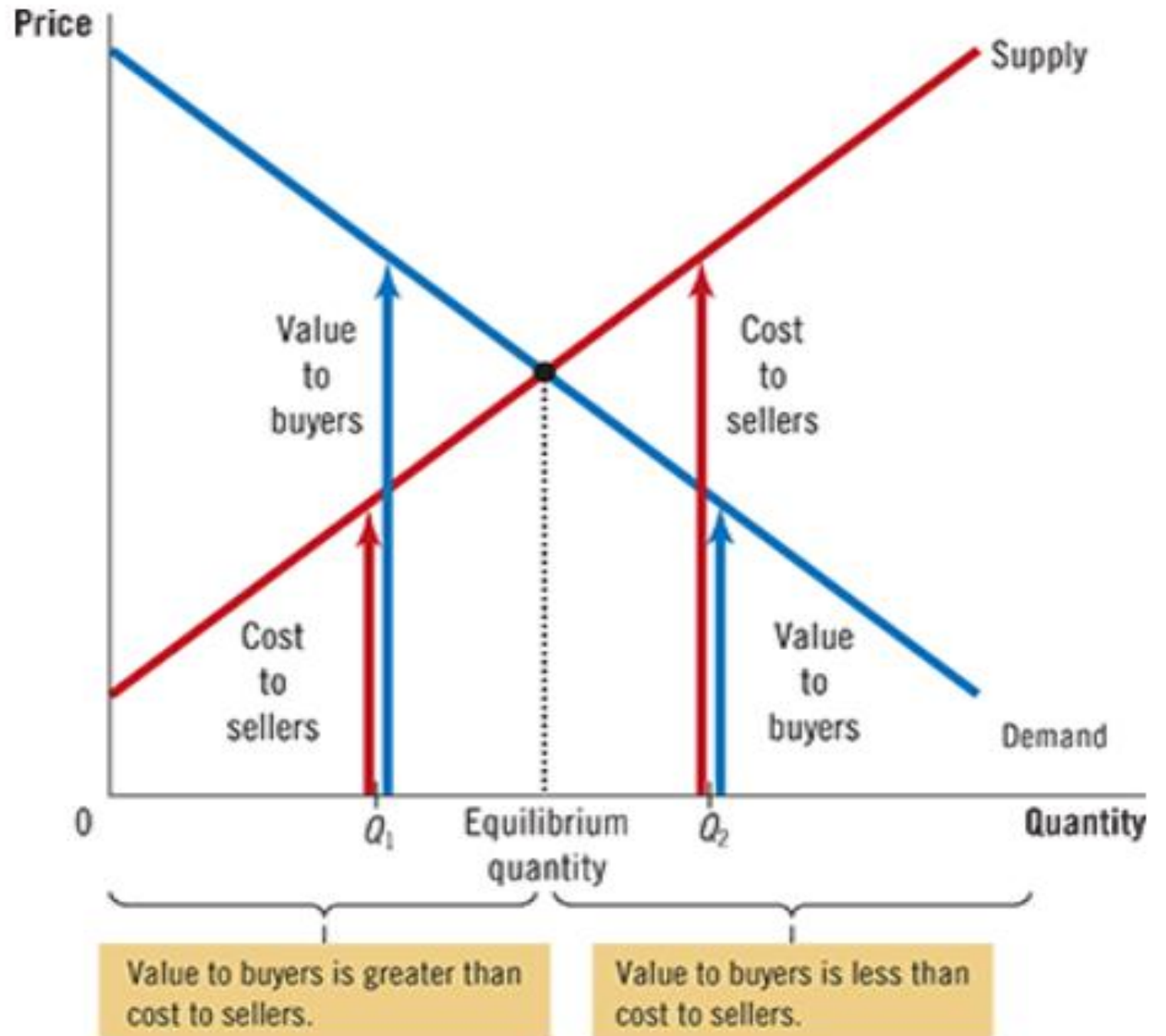
THE LAW OF SUPPLY

- Law of Supply: As price increases the Quantity Supplied increases
- This is why the Supply Curve is ALWAYS upward sloping



EQUILIBRIUM!!

- The balance (or equilibrium) between the cost to society of production and the benefit of consumption!

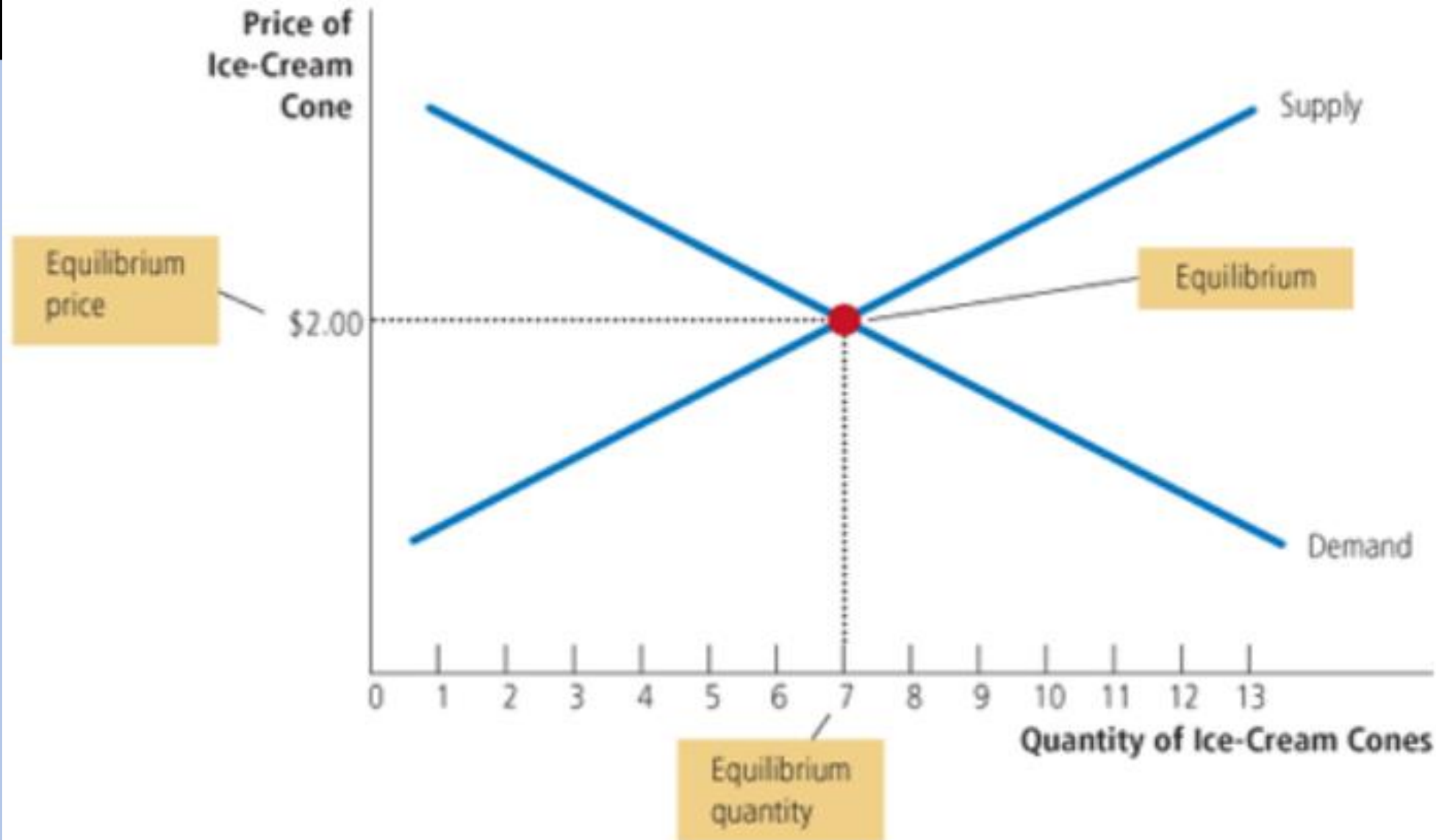


EQUILIBRIUM !!

- We take our Demand Curve and combine it with the Supply Curve to get the graph below

\$2.00 is the
Equilibrium Price

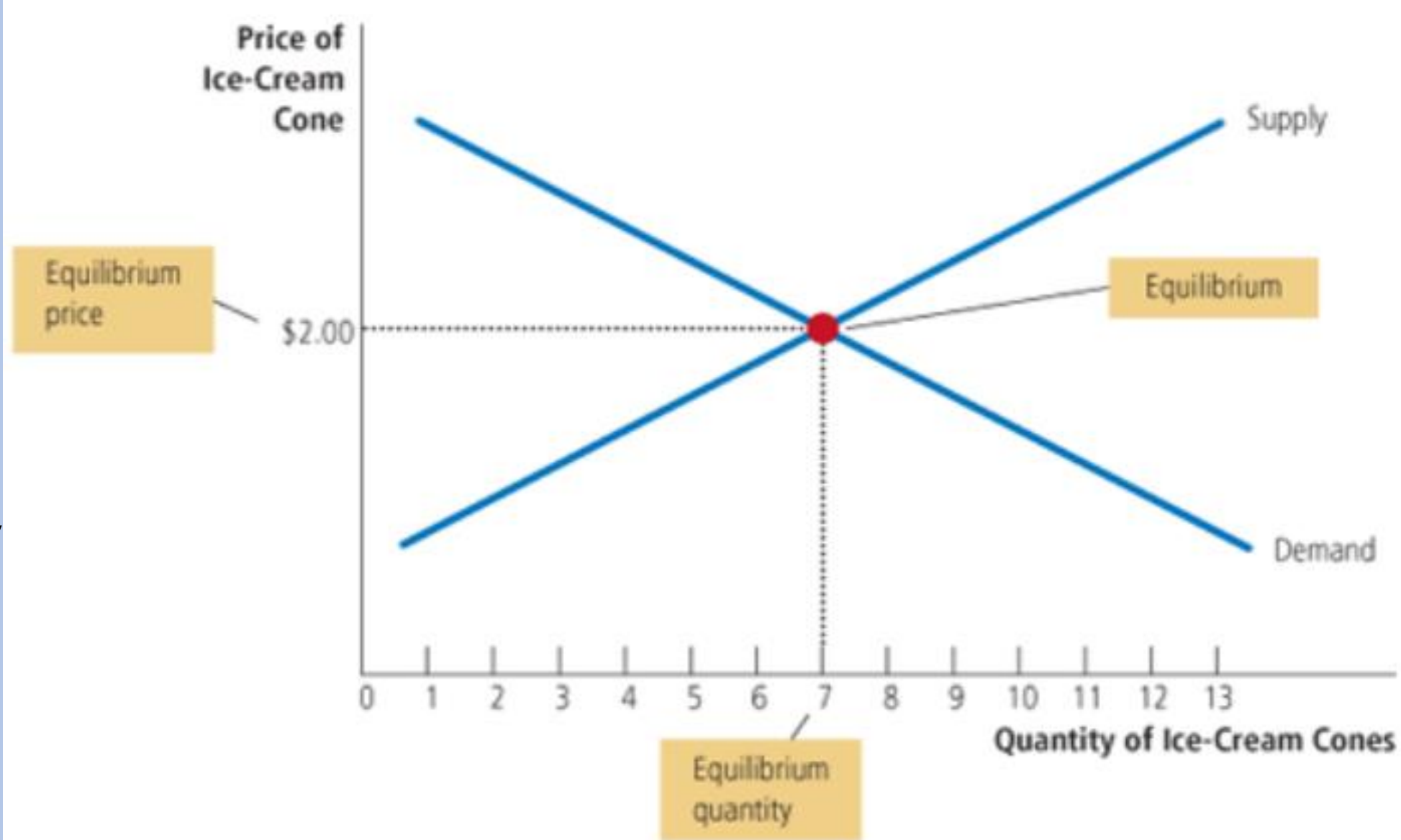
7 is the
Equilibrium Quantity



EQUILIBRIUM !!

\$2.00 is the
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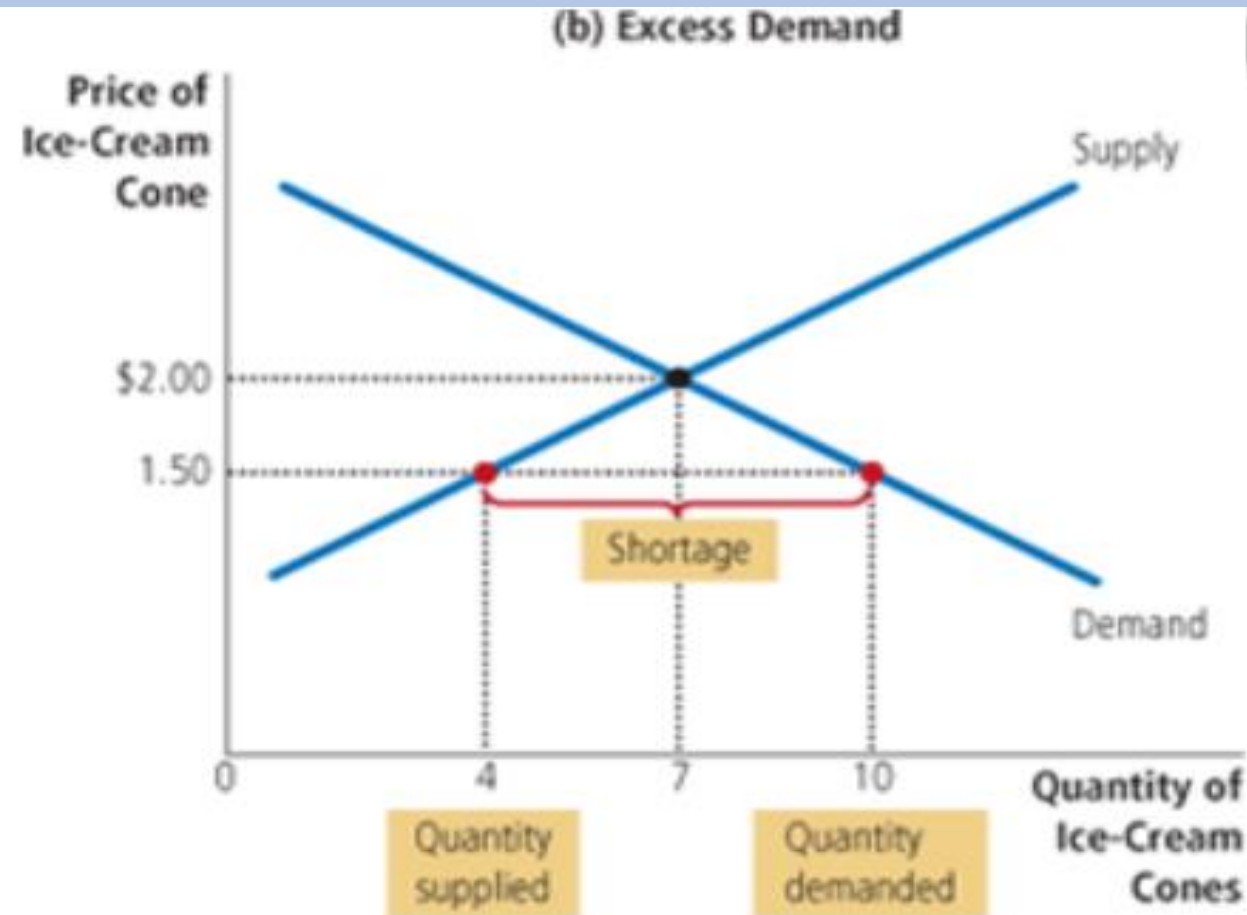
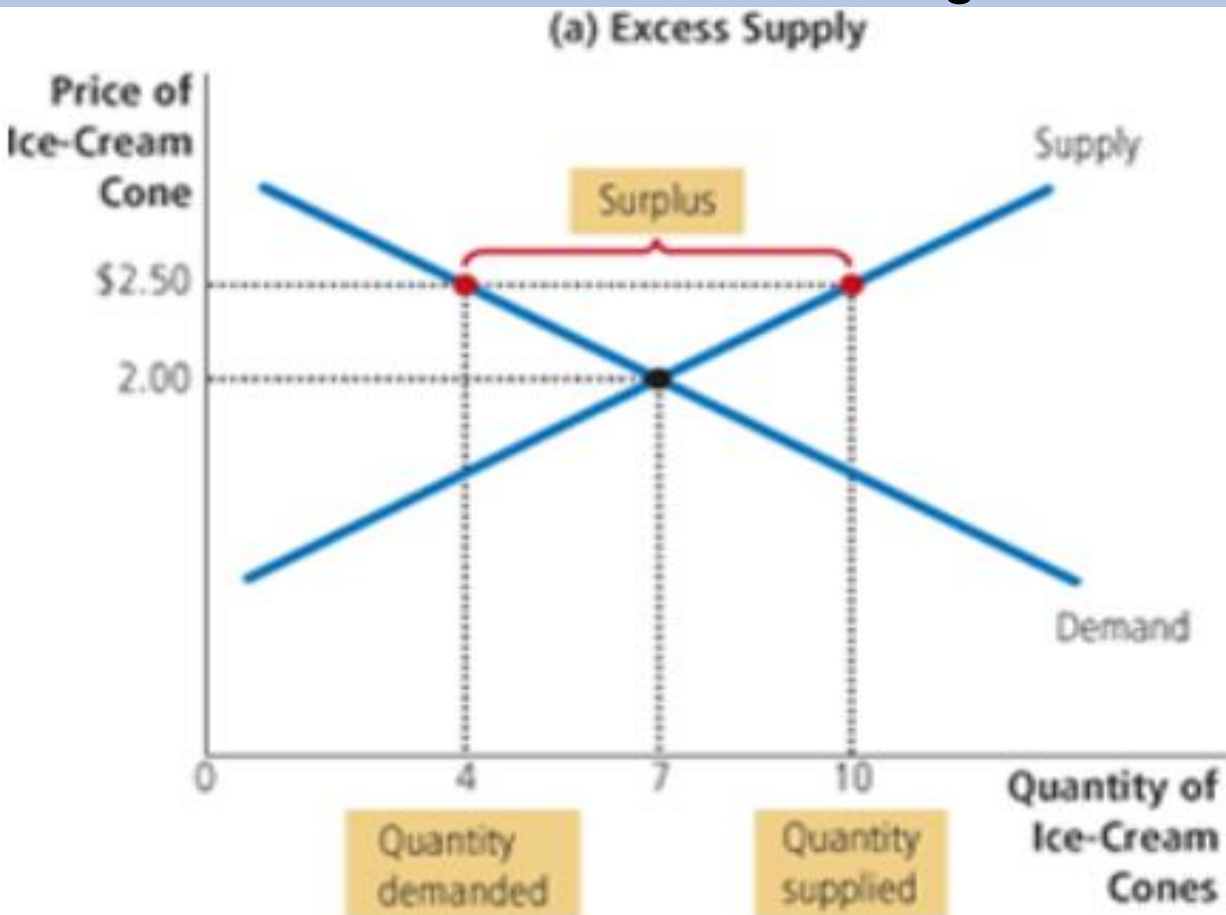
7 is the
Equilibrium Quantity



Equilibrium occurs where Quantity Supplied Equals Quantity Demanded
-Where the Supply curve and the Demand curve meet

EQUILIBRIUM !!

- When the price is too high, more firms supply the good than consumers demand of the good: the result is a surplus
- When the price is too low, more of the good is demanded by consumers than firms want to supply at that price: the result is a shortage



SUPPLY
=
DEMAND

- ONLY the EQUILIBRIUM PRICE can clear or satisfy the market
- If price is too low then more units will be demanded than supplied
- If price is too high then more units will be supplied than are demanded
- Only the equilibrium price can ensure Quantity Supplied = Quantity Demanded

Price	Quantity Demanded	Quantity Supplied	Excess Demand or Supply	Effect on Price
\$1.50	10	4	Shortage (of 6 units)	Price Must Rise
\$2.00	7	7	No Shortage or Surplus	Equilibrium Price!!!
\$2.50	4	10	Surplus (of 6 units)	Price Must Fall

A FEW TERMS

- **Change in Quantity Demanded**- A movement **ALONG** the Demand Curve caused by changes in the price of the good
- **Change in Demand**- A **SHIFT** of the **ENTIRE** Demand Curve, caused by changes in things other than price
- **Change in Quantity Supplied**- A movement **ALONG** the Supply Curve caused by changes in the price of the good
- **Change in Supply**- A **SHIFT** of the **ENTIRE** Supply Curve, caused by changes in things other than price

WHAT CAN SHIFT THE DEMAND CURVE? (CH4-2C)

- 1) Change in prices of Related Goods
- 2) Change in Tastes and Preferences
- 3) Change in Consumers Income
- 4) Change in the Number of Buyers
- 5) Change in Expectations of Price

WHAT CAN SHIFT THE SUPPLY CURVE?(CH4-3C)

- 1) Change in Costs of Inputs
- 2) Change in Technology
- 3) Change in the number of Suppliers
- 4) Expectations of Future

CHANGE IN INCOME

- An Increase in Income:
 - Rightward shift for Normal Goods
 - Leftward shift for Inferior Goods
- A Decrease in Income:
 - Leftward shift for Normal Goods
 - Rightward shift for Inferior Goods

• Normal Goods

- Things we buy more of when our incomes increase



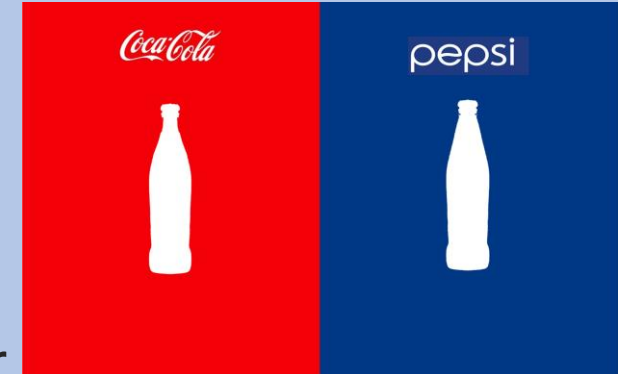
• Inferior Goods

- Things we buy less of when our incomes increase



RELATED GOODS

- Substitutes—Two goods which do the same job or are similar in another way so they can be substituted for each other in consumption
 - Ex. T-shirts and Polo shirts; Apples and Oranges; Water and Soda



- Complements—Two goods which must be used together
 - Ex. Chips and Salsa; Coffee and Milk; Peanut Butter & Jelly



RELATED GOODS

- Substitutes—If the price of a substitutes rises what should happen?
 - Example:
 - The price of Soda rises
 - Soda is more expensive so people will substitute into water
 - Demand for water will increase
- Complements—If the price of a complement rises what should happen?
 - Example:
 - The price of chips rises
 - People buy fewer chips, and thus they need less salsa
 - Demand for salsa will fall

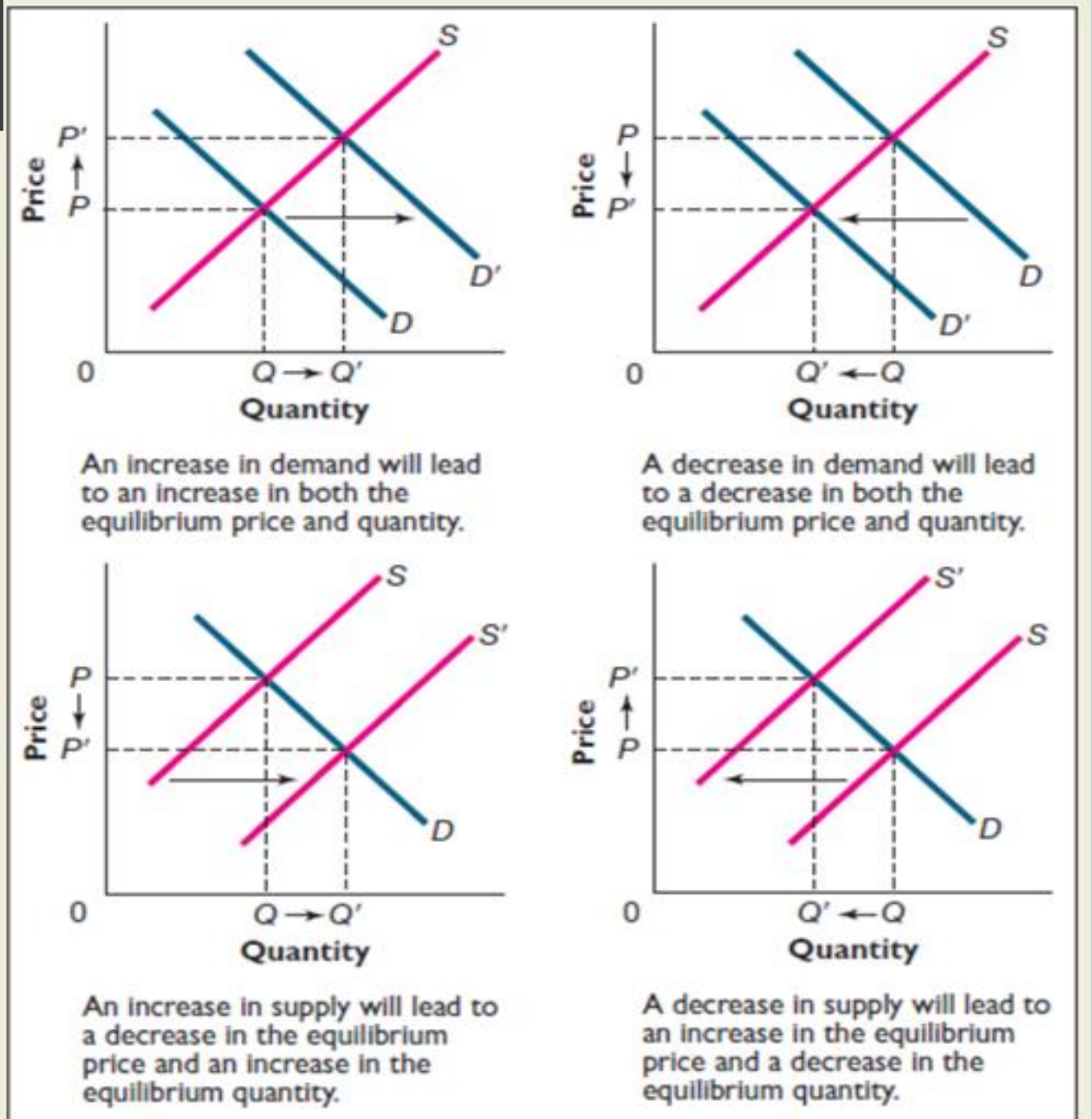
ANALYZING SHIFTS IN SUPPLY AND DEMAND

Increase in Demand Causes: **Increase in Supply Causes:**

-Increase in Equilibrium Price & Equilibrium Quantity
Increase in Equilibrium Quantity

Decrease In Demand Causes: **Decrease in Supply Causes:**

-Decrease in Equilibrium Price & Equilibrium Quantity
Decrease in Equilibrium Quantity



ANALYZING SHIFTS IN SUPPLY AND DEMAND: A CHEAT SHEET

**Increase in
Demand Causes:**

**Increase in
Equilibrium Price
&**

**Increase in
Equilibrium
Quantity**



ANALYZING SHIFTS IN SUPPLY AND DEMAND: A CHEAT SHEET

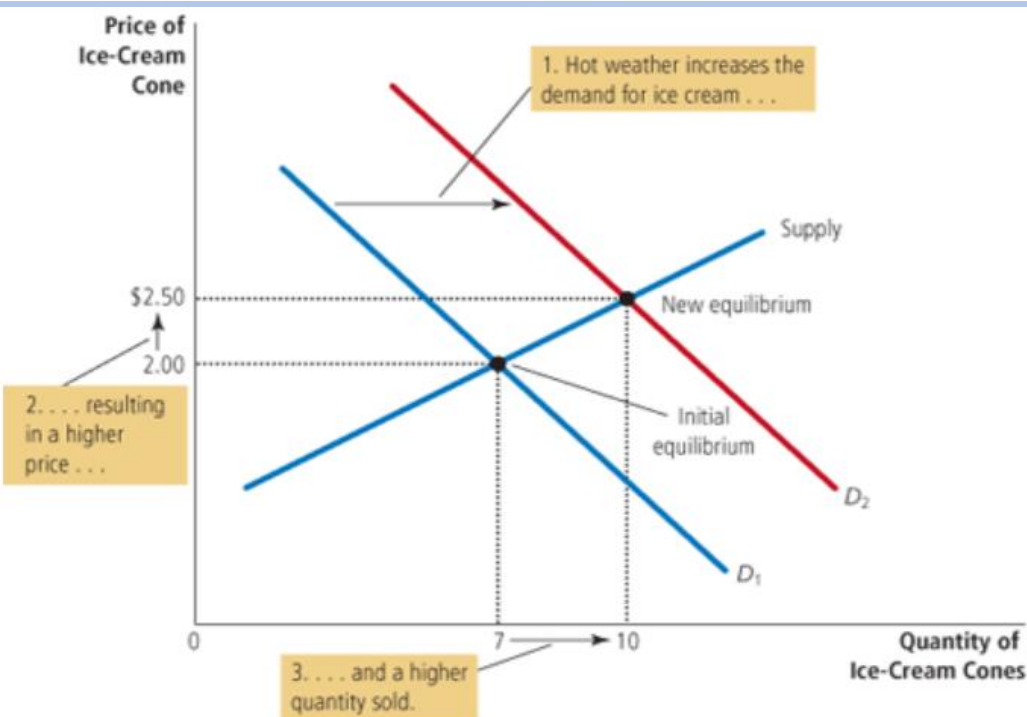
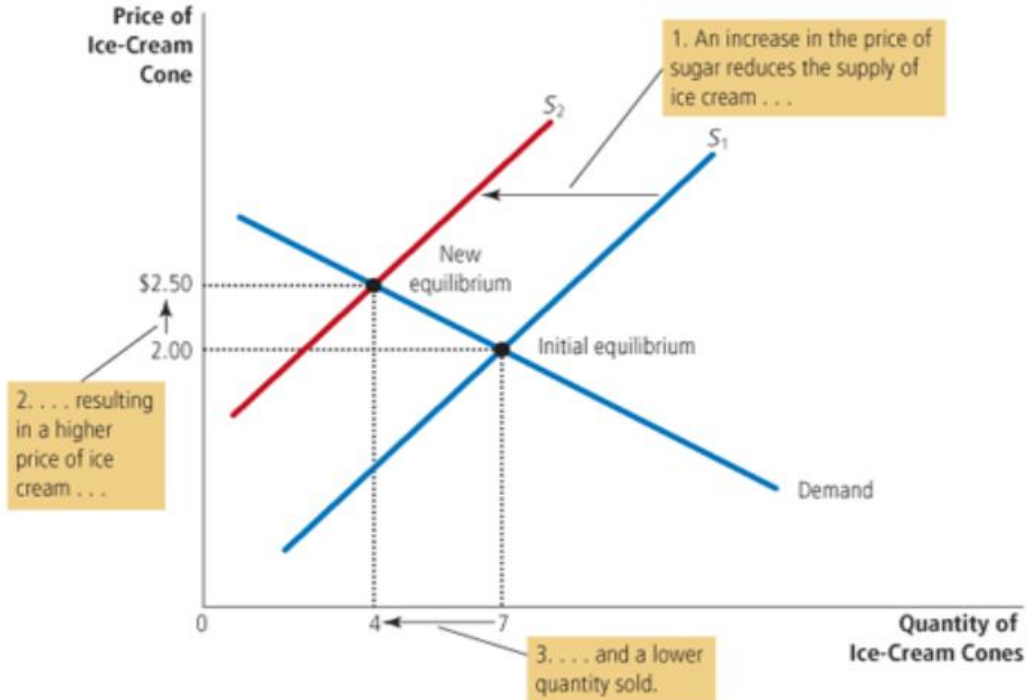
**Decrease in
Demand Causes:**

**Decrease in
Equilibrium Price
&**

**Decrease in
Equilibrium
Quantity**

SIMULTANEOUS SHIFT PROBLEMS

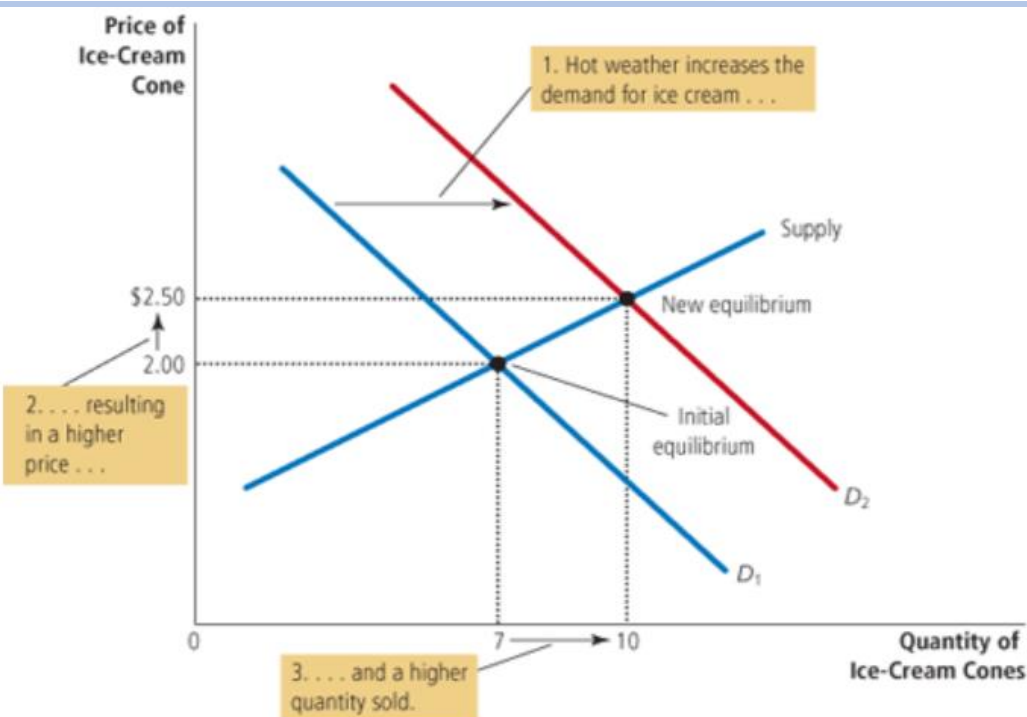
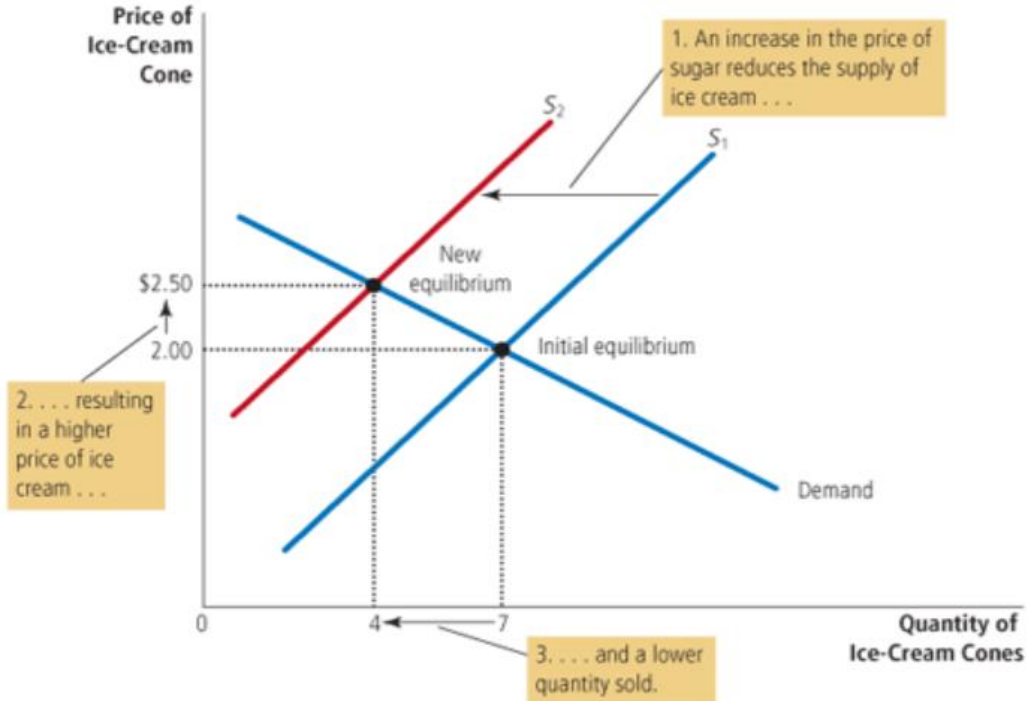
- Step 1: Draw out each Supply and Demand graph
- **The top graph is a Decrease in Supply alone**
 - Causes an increase in equilibrium price and a decrease in equilibrium quantity
- **The bottom graph is an Increase in Demand alone**
 - Causes an increase in equilibrium price and an increase in equilibrium quantity



EX: DECREASE IN SUPPLY & INCREASE IN DEMAND

Step 2:

- Decrease in Supply: Increases Equilibrium Price & Decreases Equilibrium Quantity
- Increase in Demand: Increases Equilibrium Price & Increases Equilibrium Quantity
- Both a Decrease in Supply and an Increase in Demand increase equilibrium price
- So we know, no matter what, when both shifts take place at the same time, PRICE WILL INCREASE
- But a Decrease in Supply decreases quantity and an increase in Demand increases quantity



ELASTICITIES

ELASTICITY

- Measures the responsiveness of demand to changes in:

- Changes in the price of the good

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

- Changes in the income of consumers

$$\text{Income elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

- Changes in the prices of a related good

$$\text{Cross-price elasticity of demand} = \frac{\text{Percentage change in quantity demanded of good 1}}{\text{Percentage change in the price of good 2}}$$

PRICE ELASTICITY

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

- **Elastic** goods has a **Price Elasticity of Demand** greater than 1
 - These are goods for which, when the price changes, we change our quantity demanded a lot
 - See Ch5-2d



More likely to be elastic if:

- Goods with lots of substitutes
- Luxuries
- Longer Time Horizon

PRICE ELASTICITY

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

- **Inelastic** goods has a **Price Elasticity of Demand less than 1**
 - These are goods for which, when the price changes, we change our quantity demanded by very little
 - See Ch5-2d

Shorter Time Horizon

No (or few) Substitutes



Necessities



NORMAL VS INFERIOR GOODS

- Normal Goods
 - Things we buy more of when our incomes increase
- Inferior Goods
 - Things we buy less of when our incomes increase



INCOME ELASTICITY

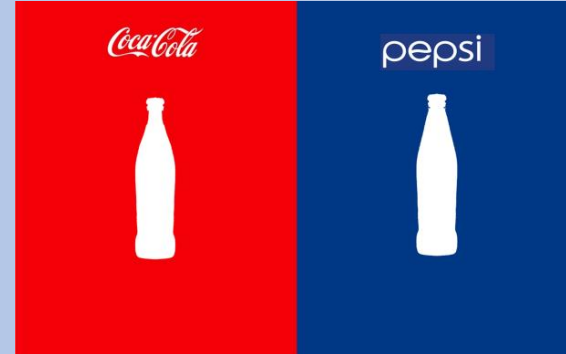
$$\text{Income elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$$

- Measures the responsiveness of Demand to changes in consumer income.
- But consumer demand will respond differently depending on if the good is a:
 - **Normal Goods**
 - Buy more when our incomes increase
 - Positive Income Elasticity of Demand
 - **Inferior Goods**
 - Things we buy less of when our incomes increase
 - Negative Income Elasticity of Demand



RELATED GOODS

- **Substitutes**—Two goods which do the same job or are similar in another way so they can be substituted for each other in consumption
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- **Complements**—Two goods which must be used together
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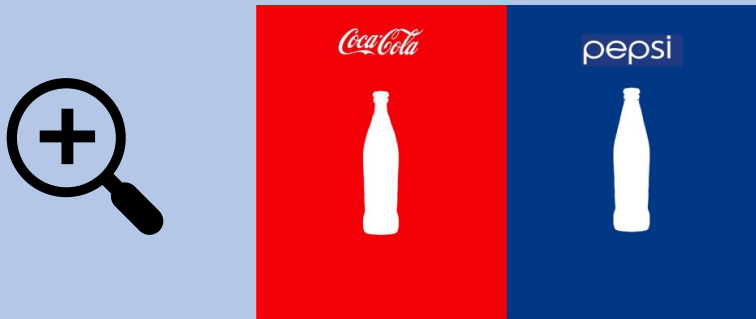
CROSS-PRICE ELASTICITY

$$\text{Cross-price elasticity of demand} = \frac{\text{Percentage change in quantity demanded of good 1}}{\text{Percentage change in the price of good 2}}$$

- Cross-Price Elasticity measures the responsiveness of demand to changes in the price of a related good.

• Substitutes

- Buy more of the substitute when price increases
- Positive Cross-Price Elasticity of Demand



• Complements

- If the price of one goes up, we buy less of both
- Negative Cross-Price Elasticity of Demand



SUPPLY



- Elasticity can work the same way for Supply

- Price Elasticity of Supply for example measures how much Quantity Supplied responds to a change in Price
- For example, housing supply is inelastic, so even when price changes the quantity supplied stays the same (in the near term).
- Or if it is the Supply of potato chips, it would be relatively easy to increase Supply in response to a price increase.